UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION 1 (5 marks)

The evolution of computers and technological advancements have led to powerful variations that can process data faster, perform complex calculations, support artificial intelligence, enable global communication, manage vast databases, simulate real-world scenarios, and offer automation, all while being smaller, more energy-efficient, and widely accessible.

a) Name the primary technology in the zero, first, and second computer generations.

b) List TWO (2) fundamental components of integrated circuits.

QUESTION 2 (5 marks)

Henry Edward "Ed" Roberts
(September 13, 1941 -April 1, 2010)
was an American engineer, entrepreneur and medical doctor who invented the first commercially successful personal computer in 1975. He is most often known as "the father of the personal computer". The computer was available as a kit for $439 or assembled for $621 and had several additional add-ons such as a memory board and interface boards. By August 1975, over 5,000 Altair 8800 personal computers were sold; starting the personal computer revolution.

a) In which computer generation that a personal computer is introduced?

b) Discuss ONE (1) other major discovery that is found in this generation.

c) Advances in laptop and hand-held devices are accelerating during this computer generation. How such technology is feasible as compared to previous computer generation?

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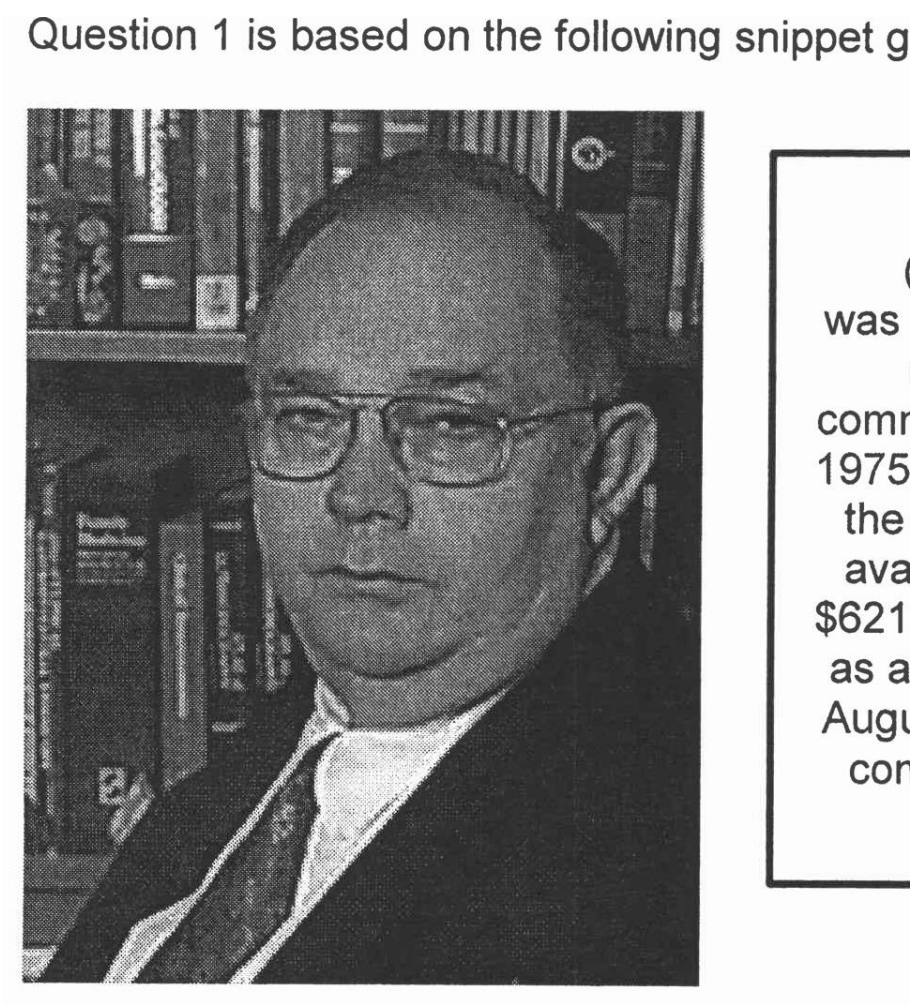


Figure 1: Snippet from a computer science web page.

QUESTION 3 (5 marks)

a) Solve 246s - 1Ai6 using the sign and magnitude method. The answer must be in byte form. Show the workflow.

b) Calculate -3Ei6 + 45s using 1's and 2's complement method. The answer must be in byte form. Show the workflow.

QUESTION 4 (5 marks)

a) List THREE (3) interfaces to external device.

b) Discuss 'hot connection' of device.

QUESTION 5 (10 marks)

a) Draw Single Instruction, Single data (SISD) architecture with the following information:
- CU is the control unit
- PU is the processing unit
- MU is the memory unit
- IS is the instruction stream

- DS is the data stream

b) List TWO (2) examples of shared memory Multiple Instruction Multiple Data (MIMD).

c) Discuss the difference between what happens if a virus attacks shared memory Multiple Instruction Multiple Data (MIMD) and distributed memory Multiple Instruction Multiple Data (MIMD).

d) Explain ONE (1) benefit of using distributed memory Multiple Instruction, Multiple Data stream (MIMD) architecture with example.

QUESTION 6 (10 marks)

a) Given the architecture of Single Instruction, Single Data Stream (SISD) is in Figure 2, with
CU representing Control Unit, PU representing the Processing Unit, and MU representing the Memory Unit. Draw the following multiple processor organization with an appropriate label. Clearly show the Instruction Stream (IS) and Data Stream (DS).

i. Multiple Instructions Multiple Data streams (MIMD) with shared memory,

ii. Multiple Instructions Multiple Data streams (MIMD) with distributed memory.

b) Give ONE (1) example for the following multiple-processor organization.

i. Multiple Instructions Multiple Data streams (MIMD) with shared memory,

ii. Multiple Instructions Multiple Data streams (MIMD) with distributed memory.

c) Discuss ONE (1) advantage of using Multiple Instructions Multiple Data stream (MIMD) with distributed memory.

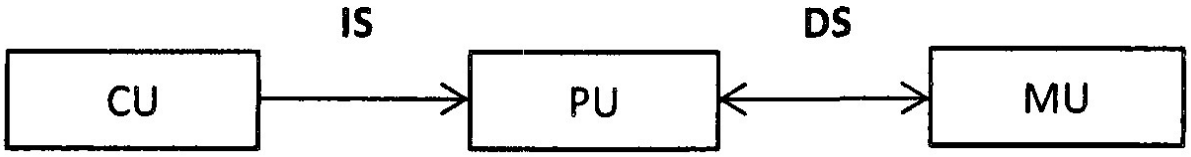


Figure 2: Single Instruction, Single Data Stream (SISD)

QUESTION 7 (10 marks)

Describe the function of each register in the computer system.

QUESTION 8 (10 marks)

Describe FOUR (4) type of access methods based on hardware implementation of the storage device. Provide ONE (1) appropriate example for each access type.

QUESTION 9 (10 marks)

a) Convert the following values to the respective number bases. Show the complete work flow.

i. 10111.1012 to decimal

ii. 507.87510 to hexadecimal

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b) Calculate 98816 - ABIO using the following techniques. Show the complete work flow.

i. Sign and magnitude

ii. 2's complement

QUESTION 10 (10 marks)

a) Discuss the function of each Central Processing Unit (CPU) component'

b) Elaborate on what will happen in the machine cycle if the bus system is faulty

PART B (20 MARKS)

Answer the question in this part.

QUESTION 11 (20 marks)

Question 1 is based on Figure 3 and Table 1.
write an appropriate program using correct assembly language notations to compute the equation snbwn in Figure 3 using instructions for zero-address, one-address, and three-address. The available instructioni are provided in Tabte 1. The register labeled A, B, C, and D are fixed, white registers x, y, and z can be used temporarily. The final answer must be stored in register z.

$$z = A^2 - B * C + \frac{2}{D + 5}$$

Figure 3

Table 1: Machine instruction

Zero-address	One-address	Three-address
PUSH p	LOAD p	ADD (p,q,r)
POP p	STOR p	suB (p,q,r)
ADD	ADD p	MUL (p,q,r)
SUB	SUB p	DIV (p,q,r)
MUL	MUL p	
DIV	DIV p	

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